

A Budget-Gated Reliability Stack for Limit-Aware In-Car Agents: CAR-bench Track 2 with an Open 120B Model

Maximilian Stephan¹, Simon Franken²

¹University of Augsburg

²Independent

maximilian.stephan@uni-a.de, mail@simon-franken.de

Abstract

We describe our Track 2 submission to the CAR-bench competition: an agent built on the open-weight `gpt-oss-120b` model served by Cerebras. Instead of scaling the model, we spend the track’s inference budget on *reliability*. A three-stage pipeline — constrained generation with deterministic enforcement retries, a risk-gated LLM verifier, and a bounded revision — is hard-capped in code at the track limit of five sequential LLM calls per baseline step, with the cap enforced by a shared per-step budget rather than by convention. Three deterministic guards add reliability at *zero* additional calls: phase separation (gather before acting), literal argument traceability, and a capability-catalog diff that tells the agent which ordinarily available tools are absent from the current session. On the public test split our agent reaches 58.7% Pass³, up from 43.3% for our prompt-only baseline, placing an open 120B model in the range reported for substantially larger proprietary systems. We report ablations, four techniques that *failed*, and the measurement noise floor that makes such comparisons fragile.

1 Introduction

CAR-bench evaluates whether an in-car assistant stays *consistent* and *limit-aware* under real-world uncertainty [1]. Its three task families probe distinct failure modes: `BASE` measures ordinary task completion, `HALLUCINATION` removes tools from the session and checks whether the agent admits the limitation instead of fabricating a result, and `DISAMBIGUATION` checks whether ambiguity is resolved from context or, when it cannot be, surfaced to the user. Scoring is strict: a task counts only if tool use, final state, intermediate state, and policy compliance are all correct, and we report Pass³ (all three trials of a task succeed), which punishes non-determinism as hard as outright error.

Track 2 constrains inference rather than model choice: at most five *sequential* LLM calls per baseline step (parallel calls inside a step are permitted) and at most 500k tokens per task on average. The reference baseline uses one call per step. This framing invites a specific question, which is the subject

of this report: *given a mid-sized open model, what is the most score-efficient way to spend four additional calls?*

Our answer is that most of the gain does not come from the extra calls at all. The largest single improvement in our system is a deterministic, zero-call prompt injection; the second largest is a deterministic validator. The extra LLM calls are worth spending only when they are *gated* — fired on a minority of steps that heuristics flag as risky. We contribute: (i) a per-step call budget that makes the track constraint a code invariant (Section 2); (ii) a set of deterministic guards derived from CAR-bench’s own error taxonomy (Section 3); (iii) an evaluation showing 43.3% $\rightarrow$ 58.7% Pass³ with a per-technique breakdown (Section 4); and (iv) an honest account of four techniques that did not survive validation, plus the noise floor that nearly hid that fact (Section 5).

2 Architecture and Call-Budget Compliance

2.1 Unit of accounting

We define one *baseline LLM step* as one action generation: one A2A round in which the agent receives a message and returns exactly one action (tool calls or a user-facing response). This is the unit the reference baseline serves with a single LLM call, and it corresponds one-to-one to an invocation of `execute()` in our agent. The budget is allocated and reset per invocation, so the constraint holds per step by construction.

We stress this because the A2A token-reporting field `turn.metrics.num_llm_calls` aggregates over a whole conversational *turn* (several rounds of tool calling before the agent finally speaks) and therefore legitimately exceeds five. Auditors should read our per-step log line `sequential_llm_calls_this_step`, which is emitted once per baseline step.

2.2 Pipeline

Figure 1 shows the pipeline. Stage ① generates a next action under a JSON response schema that forces the model to fill a `checks` object (`missing_capability`, `unspecified_value_source`, `scope_ok`) *before* naming an action, which acts as a short structured chain of thought. The parsed action then passes three deterministic validators; a violation raises a malformed-response error that triggers a retry carrying a targeted correction, but never on the

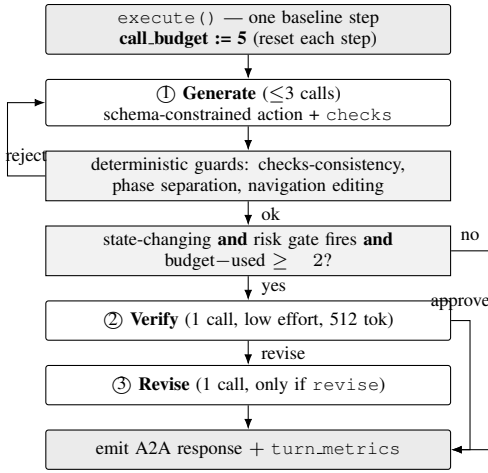


Figure 1: One baseline LLM step. All stages draw from a single per-step budget; the worst case is $3 + 1 + 1 = 5$ sequential calls, the common case is 1–3. No parallel calls are used.

Stage		Max	Total
①	Generate (1 + 2 enforcement retries)	3	3
②	Verify (gated: needs ≥ 2 left)	1	4
③	Revise (gated: budget – used)	1	5

Table 1: Worst-case sequential calls per baseline step. Measured average token use is $\sim 105k$ per task against the 500k limit.

final permitted attempt — an inconsistent action is preferable to a turn error.

Stage ② is an LLM verifier, deliberately narrow: it runs only on state-changing actions, uses low reasoning effort and a 512 token cap, and returns `approve` or `revise` with problems. A risk gate suppresses it entirely in the common, uninteresting case — a single state-changing call whose argument values are all literally traceable to the conversation. Stage ③ is a single bounded revision seeded with the verifier’s problems. Neither stage can fail a turn: on any error the stage ① action stands.

2.3 Why the bound holds

Table 1 accounts for the worst case. The key mechanism is that every stage receives the *remaining* budget, not a private allowance: stage ① caps its attempts at $\min(\text{retries} + 1, \text{budget})$; the verifier runs only if at least two calls remain (so that a revision it requests is always affordable); the revision receives exactly what is left. An optional completeness pass and a decline judge exist in the code but are disabled in the submitted configuration; both share the same guard, so the bound is independent of those toggles. Retries are counted as calls, which is conservative — a lenient reading would exclude them.

3 Reliability Techniques

Zero-call deterministic guards. Three validators inspect the parsed action and, on violation, force a corrected regeneration within stage ①. *Checks-consistency* rejects actions

that contradict the model’s own self-report (e.g. declaring a missing capability and then changing state anyway). *Phase separation* implements the recommendation from the CAR-bench analysis to separate information gathering from execution: no state change without a prior read-only call, and no clarifying question without a tool attempt since the last user message. *Navigation editing* forces the agent to observe navigation state before replacing an active route — exploiting a property of the reward function, namely that intermediate-state credit is a *subset* check over state hashes, so additional read-only calls are free while wrong state changes are not.

Argument-level anti-hallucination. Every argument of a state-changing call must appear literally in the conversation (user utterance or tool result); values that are invented or silently defaulted are rejected. A separate validator catches template placeholders ($\{\{...\}\}$, $<...>$, `tbd`) that the model emits when chaining dependent calls. This second validator alone moved BASE from 69% to 76–78%, mostly on multi-step route lookups.

Capability-catalog notice. The agent ships a static catalog of the tools and parameters that are ordinarily present in this domain, compiled from the public training split. At prompt-construction time it diffs the catalog against the tools the evaluator actually supplied and, if anything is missing, injects a notice naming the gap and instructing the agent to say plainly that it cannot do that part — and not to ask the user to supply it, claim success, or fake a workaround. This is a static prompt prior, not runtime introspection: the diff consumes only evaluator-provided tool definitions, and the resulting behaviour stays fully visible to the hallucination metric. It is our single largest win (HALLUCINATION 42% $\rightarrow$ 52–54%, twice confirmed) and fired zero times across all 75 BASE and DIS-AMBIGUATION tasks, i.e. it is inert where it should be. Two smaller injections follow the same pattern: a toll-disclosure reminder when a retrieved route sets `includes_toll`, and a notice not to re-ask for information the user has just declined to provide.

Gated verification. Over a full run the verifier issued 86 reviews and requested revision in 21% of them; inspection of a sample showed substantive catches (a policy violation, invented values, unrequested extra actions) rather than stylistic nagging. Gating matters: the full stack costs only +13% LLM time versus the single-call baseline, because most steps never reach stage ②.

4 Results

Table 2 reports full runs on the public test split (125 tasks, three trials, temperature 0). The stack raises Pass³ from 43.3% to 58.7%, with the best configuration confirmed twice in the 56–59% range. For context, the published CAR-bench leaderboard places the same open model at 28% and reports 41% for `gemini-2.5-flash-thinking` and 54% for `GPT-5-thinking`; our figures are not directly comparable (different split conditions and user-simulator routing), but the gap between 28% and 58.7% for one and the same open model is the result we consider meaningful.

The per-family pattern is informative. DISAMBIGUATION responds to structural interventions (phase separation fired

Configuration	Base	Hall.	Dis.	Pass ³
Prompt-only baseline	60	42	28	43.3
+ guards, gated verifier	64	42	48	51.3
+ catalog notice	69	52	44	52.0
+ placeholder, toll	76	67	56	58.7
<i>confirmation run</i>	78	67	52	56.7

Table 2: Pass³ (%) on the public test split, 125 tasks $\times$ 3 trials, temperature 0. Rows are cumulative configurations, each measured in a separate full run.

170 times in one run and lifted the family 28 $\rightarrow$ 48), whereas HALLUCINATION responds almost exclusively to the catalog notice. The two families also interfere: a preference-consultation rule that solved several DISAMBIGUATION tasks eroded HALLUCINATION consistency, an instance of the completion-versus-compliance tension the CAR-bench authors describe.

5 Negative Results and Measurement Noise

We report four techniques that looked promising and did not survive. (i) A *workflow memory* mined from the training split (canonical tool prerequisites, 312 tokens) scored 73% on a focused slice but 48.0% on a full run versus 51.3%; the prime suspect is a “let the user choose” phrasing that induced asking in tasks requiring internal resolution. (ii) A *response scaffold* forcing an explicit cannot-statement plus fallback offer produced the first-ever full flips on our stubborn missing-response class in focused runs, yet that class flipped back at the run level despite 88 activations; after seven attempts we closed it as a model-capability limit. (iii) A *completeness verifier* raised Pass@3 to 80% while Pass³ fell from 52.0% to 44.7%: 109 self-second-guessing revisions turned thirteen stable tasks flaky — a reminder that under Pass³, variance is a first-class cost. (iv) An *enrichment rule* for user preferences gained one task and broke our most stable marker task.

These conclusions are only possible because we first measured the noise floor. At temperature 0 the three trials of a run are near-clones, so borderline tasks flip at the *run* level rather than the trial level; an identical configuration measured twice on a 15-task slice produced 66% and 46%. We therefore accept a change only when it is deterministic (a $0/x \rightarrow x/x$ flip) or reproduces across at least two independent full runs, and we treat ± 8 pp on DISAMBIGUATION as noise. A second, non-obvious confound: the evaluator’s user simulator is itself an LLM, and provider congestion truncated its JSON badly enough to produce a 18.7% run that had nothing to do with agent quality. Run-level evaluator error counts correlate with score and must be checked before any comparison.

Where the remaining errors live. To decide when to stop, we classified the deterministically failing tasks of our best run by root cause. Only about four of thirty were cleanly addressable by harness work. The remainder split into route or destination selection (5 tasks, a reasoning failure), under-acting and omitted actions (4, semantics), final state diverging from ground truth (4, reasoning), navigation state that is simply

Enabled	MAX_CALLS_PER_STEP=5, VERIFIER, VERIFIER_GATE, ENFORCE_CHECKS, PHASE_SEPARATION, NAV_EDITING, CATALOG_NOTICE, TOLL_NOTICE
Disabled	COMPLETENESS, WORKFLOWS, RESPONSE_SCAFFOLD, DECLINE_JUDGE
Sampling	temperature 0, 2048 completion tokens, 2 enforcement retries, verifier effort low

Table 3: Submitted configuration. Every entry is an environment variable with the stated value as its default in the submitted `scenario.toml`, so the configuration is reproduced even when nothing is injected but the API key.

not observable unless the agent chooses to look (5), and content or formatting requirements such as toll disclosure and 24-hour times. This distribution is why we stopped adding mechanisms: the harness can enforce structure and surface facts, but it cannot supply reasoning the underlying model does not have. It also explains the asymmetry in Table 2 — interventions that convert an *observability* gap into an explicit prompt fact (the catalog notice, the navigation-state guard) pay off immediately, whereas the reasoning-bound classes barely move.

6 Compliance and Reproducibility

The submitted agent uses Cerebras-hosted `gpt-oss-120b` through the Cerebras SDK, with all model names, routes, API bases, service tiers, reasoning-effort selectors and sampling parameters exposed as environment variables in the submitted `scenario.toml`; the organizers need only supply the API key. The agent uses exclusively evaluator-provided inputs — system and policy text, transcript, tool definitions, tool results — plus the static public-domain catalog described in Section 3. It does not execute CAR-bench tools, read hidden task state or answer keys, or add private tools to the decision loop, and it reports `prompt_tokens`, `completion_tokens` and `thinking_tokens` on final responses. The image is public and digest-pinned; it was validated end-to-end against the official evaluator image via the repository’s GHCR validation path. Table 3 lists the exact submitted configuration.

Two caveats belong on the record. First, the capability catalog of Section 3 is prior knowledge: it is compiled offline from the *public* training split and shipped as a static file. It contains no task, answer or hidden-state information and is never updated at run time, but it does encode the expectation that a session normally offers a particular set of tools. Second, our development numbers were produced with the evaluator’s user simulator and policy judge routed through a third-party provider rather than the organizers’ configuration; scores are therefore internally comparable but not directly comparable to published leaderboard figures.

7 Limitations and Future Work

Our development loop iterated on a fixed public split, and we measured the resulting selection bias directly: at an early

stage of development, tasks we had been tuning against scored roughly 70% while unseen tasks of the same run scored about 43%, with unseen DISAMBIGUATION as low as 13%. We reacted by validating only on full runs and by preferring mechanisms that are deterministic and domain-general — a guard that enforces “observe before replacing an active route” encodes a domain rule, not a task — but some adaptation to the public split certainly remains, and the hidden-set result should be expected to fall below our development figure.

Three directions remain open. First, the verifier currently checks several aspects in one pass; splitting it into narrow single-aspect reviewers is affordable within the call budget and may raise its 21% revision rate to a more useful precision. Second, best-of- N sampling is attractive under Pass³ but interacts badly with provider rate windows, so it needs a scheduling story before it can be evaluated fairly. Third, our workflow-memory experiment failed for what looks like a single phrasing choice rather than a conceptual reason; mined canonical tool orderings deserve a second attempt with the offending route-selection line removed and DISAMBIGUATION validated separately.

8 Conclusion

Spending a fixed inference budget on *gated* verification and *deterministic* guards, rather than on uniformly deeper reasoning, lifted an open 120B model by more than 15 points of Pass³ on CAR-bench. The most transferable lesson is that the cheapest interventions dominated: a static capability diff and an argument validator, both costing no additional LLM calls, account for the majority of the gain, while the one technique that added *ungated* model calls actively hurt by injecting variance into a metric that punishes it. Limit-awareness, in our experience, is less a reasoning problem than a plumbing problem.

References

- [1] Johannes Kirmayr, Lukas Stappen, and Elisabeth André. CAR-bench: Evaluating the consistency and limit-awareness of LLM agents under real-world uncertainty. arXiv:2601.22027, 2026.